MATH 15200: Calculus II
Midterm 1

Thursday, October 22, 2026

Name: _____

Instructions. You have 80 minutes. No notes, books, calculators, phones, or other electronic devices are allowed. Show all of your work and justify your answers; a correct answer without supporting work may not receive full credit. Give exact answers (such as $\frac{\sqrt{2}}{3}$ or 5π), not decimal approximations.

Problem	**Points**	**Score**
1	20	
2	10	
3	10	
4	10	
5	15	
6	20	
7	15	
Total	100	

1. (20 points)

(a) (10 points) Prove directly from the ε–δ definition that

$$\lim_{x \to 3} \left(x^2 - 2x\right) = 3.$$

(b) (10 points) Define a function f by

$$f(x) = \begin{cases} 3, & x \neq 1, \\ 5, & x = 1. \end{cases}$$

Prove directly from the definition that f is not continuous at 1.

2. (10 points) Let $f(x) = 1/x$ on $[1, 2]$, and let $P = \{1, \frac{4}{3}, \frac{5}{3}, 2\}$.

(a) Compute the lower sum $L_f(P)$ and the upper sum $U_f(P)$.

(b) Use part (a) to find numbers A and B such that

$$A \leq \ln(2) \leq B,$$

and explain why these inequalities hold.

3. (10 points) For $x > 0$, let

$$F(x) = \int_x^{x^2} \sqrt{t^3 + 8}\, dt.$$

(a) Find $F'(x)$.

(b) Find an equation of the tangent line to the graph of F at $x = 1$.

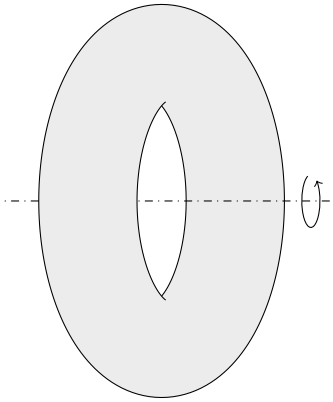
4. (10 points) Evaluate the following integrals.

(a) (5 points) $\displaystyle\int \cos x \sqrt{1+\sin x}\, dx$

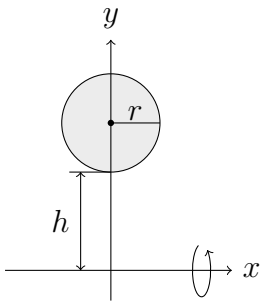
(b) (5 points) $\displaystyle\int_1^4 \frac{dx}{\sqrt{x}\left(1+\sqrt{x}\right)^2}$

5. (15 points) Find the total area of the region enclosed by the curves $y = x^3$ and $y = x^2 + 2x$.

6. (20 points) Let $r > 0$ and $h > 0$. Revolving the disk $x^2 + \big(y - (h+r)\big)^2 \le r^2$, which has radius r and center $(0, h+r)$, about the x-axis produces a solid called a *torus*; the hole in the middle of the torus has radius h (see the figure). Find the volume of the torus in terms of r and h.



the torus



the disk and the axis

Hint: Use the washer method, and note that $\int_{-r}^{r} \sqrt{r^2 - x^2}\, dx$ is the area of a semicircle of radius r.

7. (15 points) Let R be the region bounded by the curve $y = \sqrt{x}$, the x-axis, and the line $x = 4$. Find the volume of the solid obtained by revolving R about the line $x = 4$.